

Suraksha: Mobile Alarm for LPG Leakage

RoboticWares has developed an olfactory sensor-based LPG detection system that sends SMS alerts to site owners if there is gas leakage when they are away

JALAJA RAMANUNNI

Prevention is better than cure—especially to secure your home against possible fire due to LPG leakage when you are away on a holiday or work. Thankfully, to avoid the loss of lives and property due to LPG leakage, the team at RoboticWares has developed an SMS-based feedback system that sends an SMS alert to phone numbers prestored in the system as soon as its sensor detects LPG leakage. They call the device Suraksha, meaning ‘safety’ in English.

The team behind this system includes Gautam Kumar, co-founder and director of RoboticWares, co-founders Kushal Nahata and Gaurav Srivastava, and system engineers Gaurav Sonker and Ganesh Naidu.

The idea for Suraksha cropped up after Kumar read about a gas leak in Bhubaneswar that cost many lives. Nahata had memories of a similar incident in Chandni Chowk, New Delhi, a few years ago. The two felt an immediate need for developing an alert system to warn people and avoid such accidents.

How it works

The device is attached to an AC power source and can be configured to communicate with a maximum of five people by registering their mobile numbers on the device.

Users have to insert their SIM cards into the device and then add the number to Suraksha's database. In case of a gas leakage, the device rings an alarm and sends an SMS to registered users



Suraksha in 'on' position

informing them about the impending danger.

Suraksha requires a 12V, 2A power adaptor and includes an LPG gas sensor, microcontroller, global system for mobile communication (GSM) module and power circuits.

“The device uses Atmel's ATmega8 microcontroller as it has all the features we require, like analogue-to-digital conversion (ADC), universal asynchronous receiver/transmitter (UART), TWI (I²C), timers and interrupts,” shares Kumar.

Kumar adds, “The most important thing is that all the related compil-

ers and software are Open Source. For the GSM module, we initially tested Wavcom, SIM300, SIM900B and u-blox. Due to easy product availability and support, we decided to choose the u-blox module.

“Suraksha uses a GSM module and the maximum current requirement is 2A. The voltage should be in the range of 3.5V to 4.5V for all kinds of GSM modules. We required a regulator to fix the output to the required level like 3.8V, 4V or 4.2V as per the GSM module. For this, we used an LM2572/6 ADJ voltage regulator. We used 4.2V for our circuit in Suraksha.”

The team used AVR Studio and WINAVR for microcontroller programming and debugging purpose. Since they used a GSM module and the microcontroller, they did not face any trouble in interfacing. All they had to do was learn AT commands for interfacing the GSM module. They have applied for a patent for the entire LPG gas detection device with SMS alert mobile communication system.

Challenges faced

RoboticWares had earlier developed a leakage sensor with a buzzer alarm.